

DL: Md Obaidur Rahman , Ayon Sarker dada

RTL: Quazi Shafayat Hossain , Himaddri Roy

DTL:, Md. Jahin Alam

TTL: Jishnu Mahmud , Tasnim Ferdous apu

- While posting your question here, please follow this format:

(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

DL [SetA-Ques2 & SetB- Ques3]

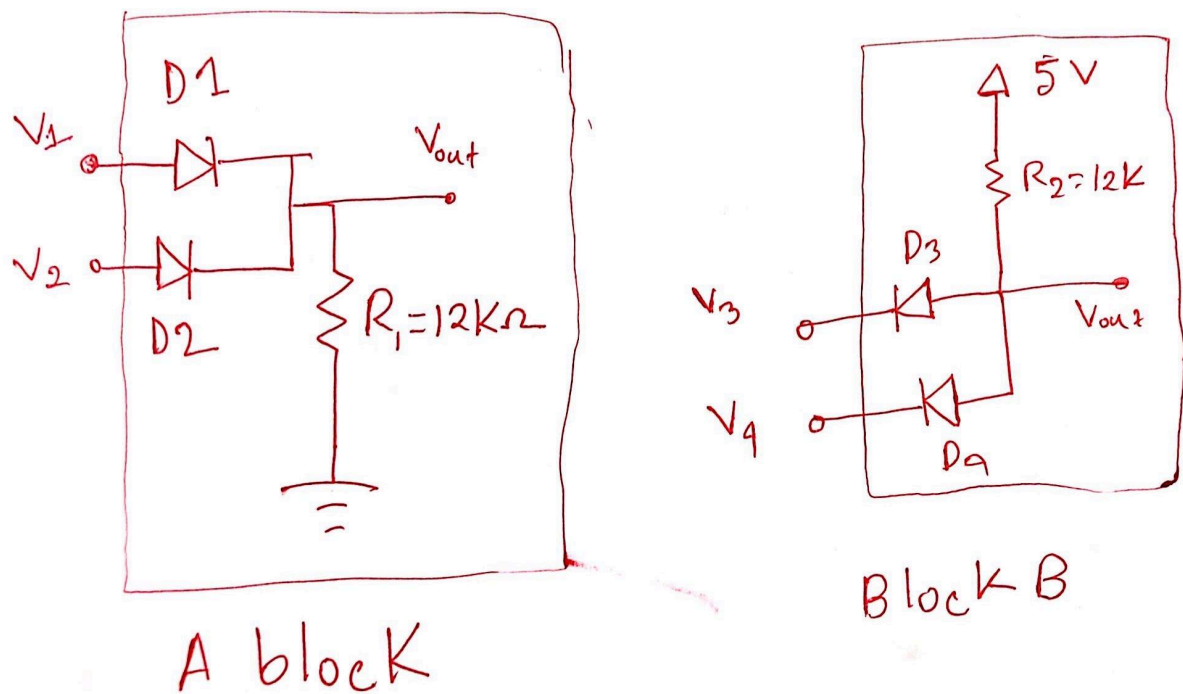


Figure :1

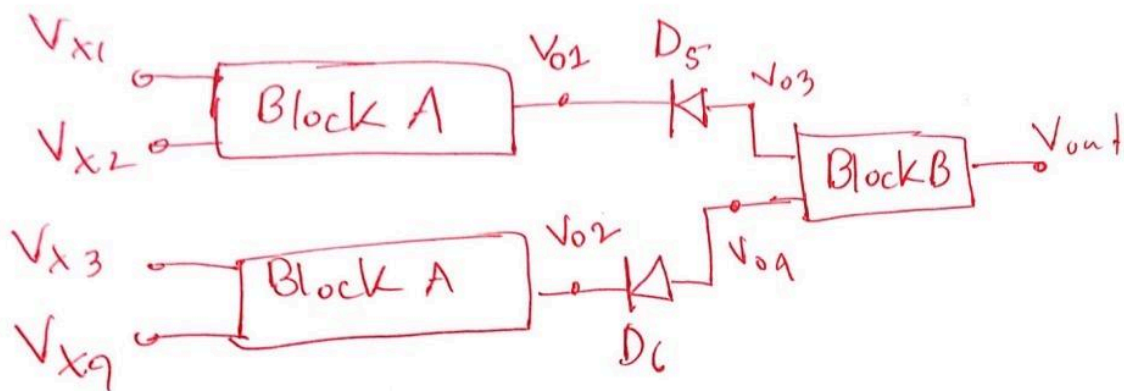


Figure :2

Block A and Block B consists of two different circuits shown in figure 1. The final circuit is constructed using these blocks and additional diodes shown in figure 2 where V_{X1} , V_{X2} , V_{X3} and V_{X4} are the inputs of the two A blocks. Consider the following two input combinations:

Case 1: $V_{X1} = 0V, V_{X2} = 0V, V_{X3} = 0V, V_{X4} = 0V$

Case 2: $V_{X1} = 5V, V_{X2} = 5V, V_{X3} = 5V, V_{X4} = 5V$

(for set B, change the resistor values to 10k)

(a)	CO1	Find the voltages at $V_{o1}, V_{o2}, V_{o3}, V_{o4}$ and V_{out} for case 2	[4]
(b)	CO1	Find the total power dissipation by each of the resistors in the overall circuit for case 1.	[4+4+4]
(c)	CO1	Implement the following Boolean function using only Block A and Block B. $V_{out} = ((V_{X1} + V_{X2}) \cdot V_{X3} \cdot V_{X4})$. (Only draw the circuit using the blocks)	[4]

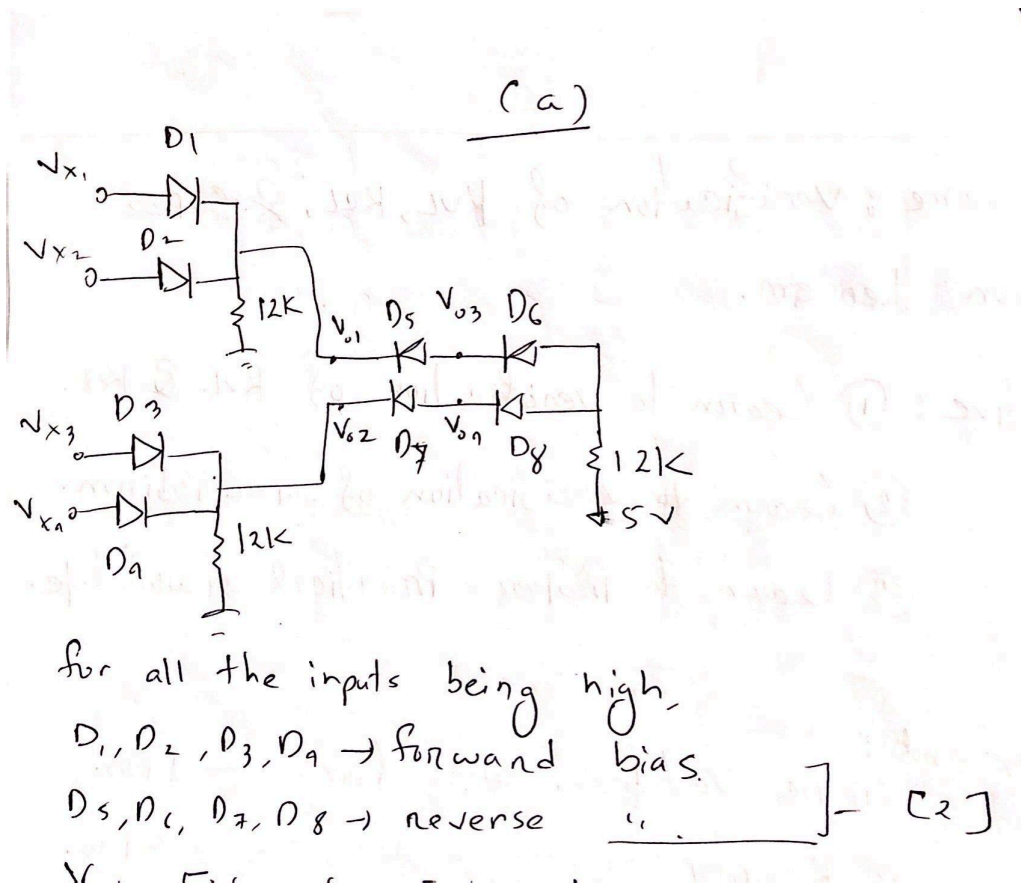
From Jishnu - Onek koshto hoise eidi latex e korte. 😊

From Himaddri - Obaidur tor khobor ache

From Obaidur - LOL.

DL Solution:

You are free to give partial marking according to your own will.



[2]

Diode cut in voltage = 0.7

$$V_{01} = 4.3V, \quad V_{02} = 4.3V$$

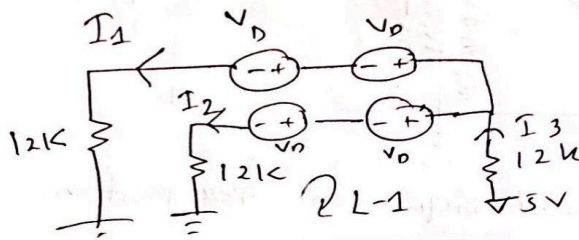
$V_{03} = 0V, \quad V_{04} = 0V$ [disconnected from both side of the circuit]
 $V_{out} = 5V$

(b)

for case 1,

$D_1, D_2, D_3, D_4 \rightarrow \text{off}$ [reverse bias]

$D_5, D_6, D_7, D_8 \rightarrow \text{on}$ [forward bias]



Applying KCL, $I_3 = I_1 + I_2$ — (i)
 Applying KVL in lower loop

$$5 - 2V_0 = 12I_3 + 12I_2$$

$$\Rightarrow \frac{3.6}{12} = I_2 + I_3 \Rightarrow I_2 + I_3 = 0.3$$
 — (ii)

Applying KVL in upper (overall) loop,

$$5 - 2V_0 = 12I_3 + 12I_1 \Rightarrow I_1 + I_3 = 0.3$$
 — (iii)

solving (i), (ii), (iii)

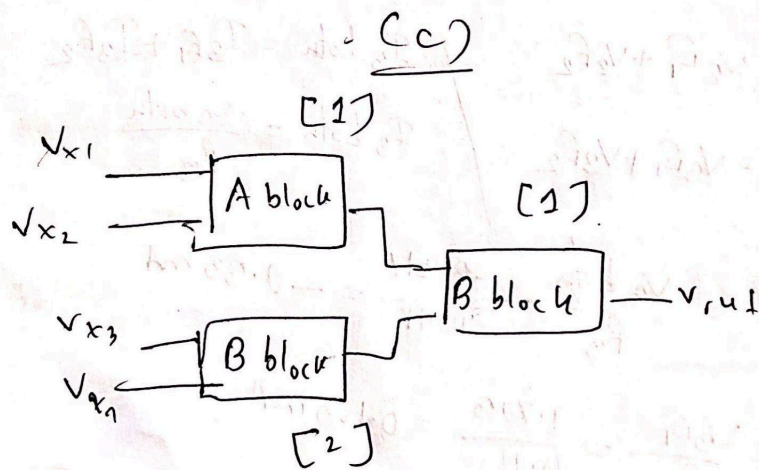
$$I_1 = 0.1mA, \quad I_2 = 0.1mA, \quad I_3 = 0.2mA$$

for set B $I_1 = 0.12mA, \quad I_2 = 0.12mA, \quad I_3 = 0.24mA$

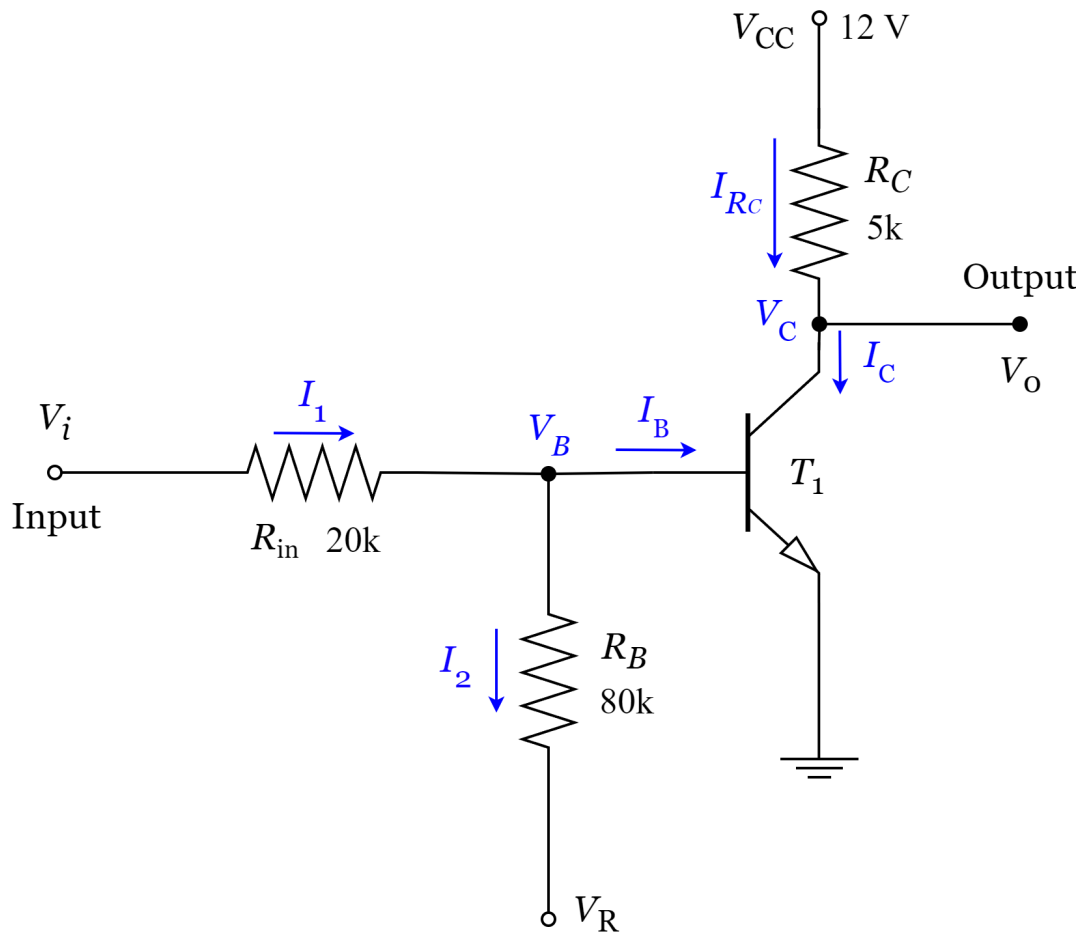
$$\begin{aligned} I_1^2 R_1 &= (0.1)^2 \times 12 = 0.12mW \\ I_2^2 R_2 &= (0.1)^2 \times 12 = 0.12mW \\ I_3^2 R_3 &= (0.2)^2 \times 12 = 0.48mW \\ \hline &0.72mW \end{aligned} \quad \left. \begin{array}{l} \text{--- (a)} \\ \text{(Ans)} \\ \text{for set A} \end{array} \right\} [4]$$

for set B,

$$\begin{array}{lcl}
 P_1 = I_1^2 R_1 = \boxed{0.144 \text{ mW}} & \text{--- [4]} & \\
 P_2 = I_2^2 R_2 = \boxed{0.144 \text{ mW}} & \text{--- [4]} & \\
 P_3 = I_3^2 R_3 = \boxed{0.576 \text{ mW}} & \text{--- [4]} & \\
 \hline
 & \boxed{0.864 \text{ mW}} &
 \end{array}
 \left| \begin{array}{l} \text{for set B} \end{array} \right.$$



RTL



(a)	CO2	Determine the value of V_R such that when $V_i = 0.2$ V, power dissipation is 1 mW. [Assume T_1 is in cutoff]	[7]
(b)	CO2	Determine the maximum value of V_i that can be applied such that transistor T_1 is in cutoff state. [Assume $V_R = -12$ V]	[7]
(c)	CO2	Determine the minimum value of β_F such that the logic gate does not malfunction. [Assume $V_R = -12$ V]	[6]

4RTL Solution: [SetA-Ques1 & SetB- Ques1]

SET A

(a) Power Dissipation = $\Delta V \times I$

$$\Rightarrow 1 \text{ mW} = (0.2 - V_R) \times \frac{0.2 - V_R}{20\text{k} + 80\text{k}}$$

$$\Rightarrow V_R = -9.8 \text{ V}$$

(b) T_1 cutoff, $V_B = \text{cut in voltage}$
 $= 0.5 \text{ V}$

$$\therefore I_1 = I_2$$

$$\Rightarrow \frac{V_i - 0.5}{20\text{k}} = \frac{0.5 + 12}{80\text{k}}$$

$$\Rightarrow V_i = 3.625 \text{ V}$$

(c) $I_B = I_1 - I_2$

$$= \frac{12 - 0.8}{20\text{k}} - \frac{0.8 + 12}{80\text{k}}$$

$$= 0.4 \text{ mA}$$

$$I_c = \frac{12 - 0.2}{5\text{k}}$$

$$= 2.36 \text{ mA}$$

$$\beta_{\min} = \frac{I_c}{I_B}$$

$$= \frac{2.36}{0.4}$$

$$= 5.9$$

SET B

(a) Power Dissipation = $\Delta V \times I$

$$\Rightarrow 1 \text{ mW} = (0.2 - V_R) \times \frac{0.2 - V_R}{18\text{k} + 90\text{k}}$$

$$\Rightarrow V_R = -10.19 \text{ V}$$

(b) T_1 saturation, on the edge of saturation and F.A.

$$I_c = \frac{10 - 0.1}{5\text{k}} \quad \left\{ \begin{array}{l} I_B = \frac{I_c}{\beta} \\ = 1.98 \text{ mA} \end{array} \right. \quad \left\{ \begin{array}{l} = \frac{1.98}{30} = 0.066 \text{ mA} \end{array} \right.$$

$$I_B = I_1 - I_2$$

$$\Rightarrow 0.066 = \frac{V_{in} - 0.8}{18\text{k}} - \frac{0.8 + 10}{90\text{k}}$$

$$\Rightarrow V_{in} = 4.15 \text{ V}$$

(c) $I_B = I_1 - I_2$

$$= \frac{10 - 0.8}{18\text{k}} - \frac{0.8 + 10}{90\text{k}}$$

$$= 0.39 \text{ mA}$$

$$I_c = \frac{10 - 0.1}{5\text{k}}$$

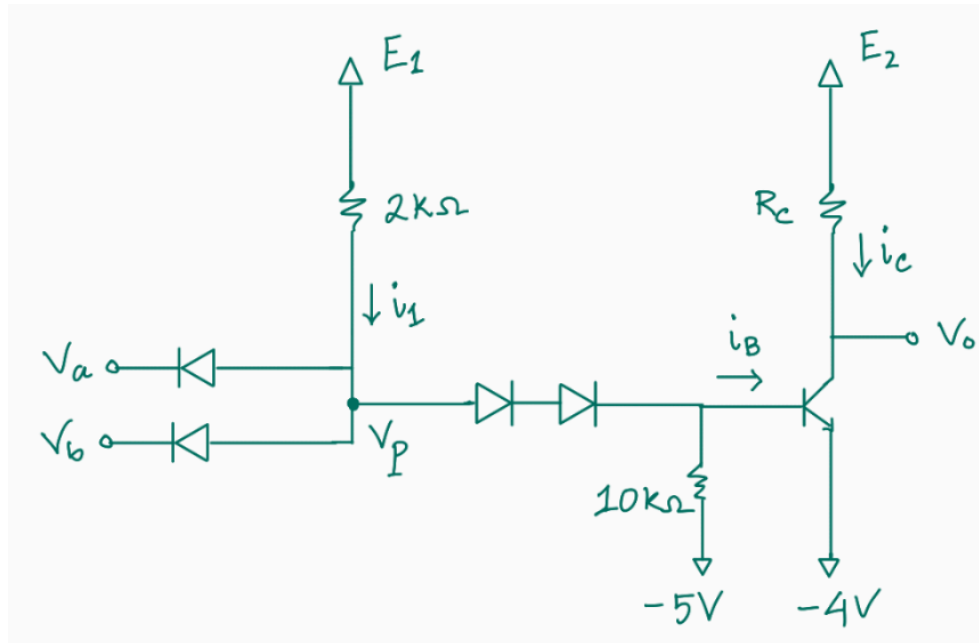
$$= 1.98 \text{ mA}$$

$$\beta_{\min} = \frac{I_c}{I_B}$$

$$= \frac{1.98}{0.39}$$

$$= 5.08$$

DTL: [MJA][SetA-Ques3 & SetB- Ques2]



In the 2-input DTL NAND Logic shown above, the input voltages V_a and V_b can be either 0.2V (Low) or 8V(High). The utilized diodes and transistor have different operating voltages than usual. Important parameters are given below:

β_F	6	V_{BE}^{Sat}	1V	V_{CE}^{Sat}	0.5V
$V_Y(BJT)$	0.8V	$V_Y(diode)$	0.6V	$V_d(diode)$	0.7V

(a)	CO1	Determine the minimum value of V_p for the transistor to operate.	[04]
(b)	CO1	Determine v_o , i_1 , i_B and i_c if both the inputs are in high state.	[08]
(c)	CO1	Is the given current gain, β_F suitable for the circuit?	[02]
(d)	CO1	What is the maximum dissipated power achievable from the circuit?	[06]

Set A: $E_1 = 8V$, $E_2 = 7V$, $R_c = 2k\Omega$

Set B: $E_1 = 7V$, $E_2 = 8V$, $R_c = 3k\Omega$

DTL Solution:

Set : A, Set : B

$$(a) \quad V_p \geq -4 + V_Y(BJT) + 2 * V_Y(diode) = -2V$$

$$(b) \quad v_0 = -4 + V_{CE}^{Sat} = -3.5V,$$

$$V_p = -4 + V_{CE}^{Sat} + 2 * V_d(diode) = -1.6V$$

$$i_1 = \frac{E_1 + 1.6}{2} = 4.8mA \parallel 4.3mA, \quad i_R = \frac{-4 + V_{BE}^{Sat} + 5}{10} = 0.2mA$$

$$i_B = i_1 - i_R = 4.6mA \parallel 4.1mA, \quad i_c = \frac{E_2 - v_0}{R_c} = 5.25mA \parallel 3.83mA$$

$$(c) \quad \beta_{min} = \frac{i_c}{i_b} = 1.14 \parallel 0.93 < \beta_F = 6. \quad [\text{Therefore, suitable}]$$

$$(d) \quad P_B = (E_1 + 4) * i_B = 55.2 \text{ mW} \parallel 45.1 \text{ mW}$$

$$P_R = (E_1 + 5) * i_R = 2.6 \text{ mW} \parallel 2.4 \text{ mW}$$

$$P_c = (E_2 + 4) * i_c = 57.75 \text{ mW} \parallel 45.96 \text{ mW}$$

$$\therefore \text{Max. dissipated power} = 115.55 \text{ mW} \parallel 93.46 \text{ mW}$$

TTL[SetA-Ques4 & SetB- Ques4]

Done directly on latex.

TTL Solution:

SET A

(a) $V_X = V_Y = \text{High}$

Q_1 Reverse Active

Q_2 Saturation

Q_0 Saturation

$$\text{Now, } I_3 = \frac{5 - 0.1}{5k} = 0.98 \text{ mA}$$

$$I_1 = \frac{5 - 2.5}{10k} = 0.25 \text{ mA}$$

$$\begin{aligned} I_{E_X} = I_{E_Y} &= \beta_R \times I_1 \\ &= 0.1 \times 0.25 = 0.025 \text{ mA} \end{aligned}$$

$$\begin{aligned} I_{B_2} &= I_1 + 2 \times I_{E_X} = 0.25 + 0.05 \\ &= 0.3 \text{ mA} \end{aligned}$$

$$I_2 = \frac{5 - 1}{4k} = 1 \text{ mA}$$

$$\begin{aligned} I_{E_2} &= I_2 + I_{B_2} = 1 + 0.3 \\ &= 1.3 \text{ mA} \end{aligned}$$

$$\begin{aligned} I_{B_0} &= I_{E_2} - I_B = 1.3 - \frac{0.9}{2k} \\ &= 0.85 \text{ mA} \end{aligned}$$

$$\begin{aligned} \text{For } Q_0, \beta_{\text{forced}} &= \frac{I_3}{I_{B_0}} \\ &= \frac{0.98}{0.85} = 1.15 < 30 \end{aligned}$$

Q_0 in saturation ~ Proved ~

$$(b) I_L = I_1^1 = \frac{5 - 1}{10k} = 0.4 \text{ mA}$$

$$\begin{aligned} \text{Now, } N &< \frac{\beta_F I_{B_0} - I_3}{I_L} \\ \Rightarrow N &< \frac{30 \times 0.85 - 0.98}{0.4} \end{aligned}$$

$$\Rightarrow N < 61.3$$

$$\therefore \text{Fanout} = 61$$

(c) I_L will be changed.

$$I_L' = I_L / 2$$

$$\therefore \text{Fanout, } N < \frac{\beta_F I_{B_0} - I_3}{I_L / 2}$$

$$\Rightarrow N < 2 \times \text{Fanout from (b)}$$

$$\Rightarrow N < 2 \times 61.3$$

$$\Rightarrow N < 122.6$$

$$\therefore \text{Fanout} = 122$$

SET B

(a) $V_X = V_Y = \text{High}$

Q_1 Reverse Active

Q_2 Saturation

Q_0 Saturation

Now, $I_3 = \frac{5 - 0.2}{5k} = 0.96 \text{ mA}$

$I_1 = \frac{5 - 2.3}{10k} = 0.27 \text{ mA}$

$I_{E_X} = I_{E_Y} = \beta_R \times I_1$
 $= 0.1 \times 0.27 = 0.027 \text{ mA}$

$I_{B_2} = I_1 + 2 \times I_{E_X} = 0.27 + 0.054$
 $= 0.324 \text{ mA}$

$I_2 = \frac{5 - 1}{4k} = 1 \text{ mA}$

$I_{E_2} = I_2 + I_{B_2} = 1 + 0.324$
 $= 1.324 \text{ mA}$

$I_{B_0} = I_{E_2} - I_B = 1.324 - \frac{0.8}{2k}$
 $= 0.924 \text{ mA}$

For Q_0 , $\beta_{\text{forced}} = \frac{I_3}{I_{B_0}}$
 $= \frac{0.96}{0.924} = 1.04 < 35$

Q_0 in Saturation ~ Proved ~

(b) $I_L = I_1^1 = \frac{5 - 1}{10k} = 0.4 \text{ mA}$

Now, $N < \frac{\beta_F I_{B_0} - I_3}{I_L}$
 $\Rightarrow N < \frac{35 \times 0.924 - 0.96}{0.4}$

$\Rightarrow N < 78.45$

$\therefore \text{Fanout} = 78$

(c) I_L will be changed.

$I_L' = I_L / 2$

$\therefore \text{Fanout}, N < \frac{\beta_F I_{B_0} - I_3}{I_L / 2}$

$\Rightarrow N < 2 \times \text{Fanout from (b)}$

$\Rightarrow N < 2 \times 78.45$

$\Rightarrow N < 156.9$

$\therefore \text{Fanout} = 156$